

Tokenization

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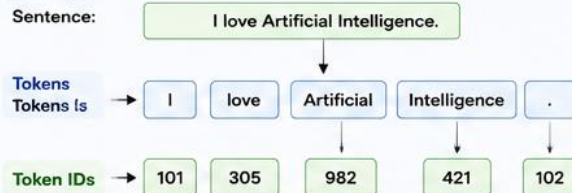
Tokenization, BPE and WordPiece – Explained in Simple Way

Before a sentence is given to a Transformer model (BERT, GPT, Llama, etc.), it is converted into tokens (subwords) and then into numbers (IDs).

1. What is Tokenization?

Tokenization is the process of breaking a sentence into smaller units called tokens. Each token is then mapped to a number (Token ID) from the vocabulary.

Example



Why Subword Tokenization?

If the model has never seen a word like “unbelievable”, it may treat it as UNKNOWN (UNK). Subword tokenization solves this by splitting it into smaller meaningful pieces.



Advantages

- ✓ Handles rare / new words
- ✓ Smaller vocabulary size
- ✓ Better generalization
- ✓ Faster and efficient

2. Byte Pair Encoding (BPE)

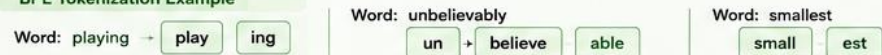
BPE starts with characters. It repeatedly merges the most frequent adjacent pair until a vocabulary of subwords is formed.

Example

Corpus: low, lowest, lower, newest

Step 1 (Start with characters)	l o w l o e s t l o w e r n e w e s t	Most frequent pair: l + o Merge → lo	lo w lo w e s t lo w e r n e w e s t
Step 2 Merge again	lo w lo w e s t lo w e r n e w e s t	Most frequent pair: lo + w Merge → low	low low e s t low e r n e w e s t
Step 3 Merge again	low low e s t low e r n e w e s t	Most frequent pair: e + s Merge → es	low low es t low e r n e w es t
Step 4 Merge again	low low es t low e r n e w es t	Most frequent pair: es + t Merge → est	low lowest lower n e w est
Final Subword Vocabulary (one possible result): low, lower, lowest, new, est, e, r, l, o, w, n, ...			

BPE Tokenization Example



3. WordPiece

WordPiece is similar to BPE but chooses merges based on the best language model probability (which merge helps predict the word better).

Example

Word: playing

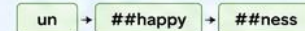
Step 1 (Start with characters)	p l a y i n g	Choose best merge (by probability): i + n	p l a y i n g
Step 2	p l a y i n g	Choose best merge: y + in	p l a y i n g
Step 3	p l a y i n g	Choose best merge: a + y in	p l a y i n g
Step 4	p l a y i n g	Choose best merge: play + in	play i n g
Step 5 (final)	play i n g	Choose best merge: in + g	play ing

WordPiece Tokenization Examples

Word: playing



Word: unhappiness



Word: smallest

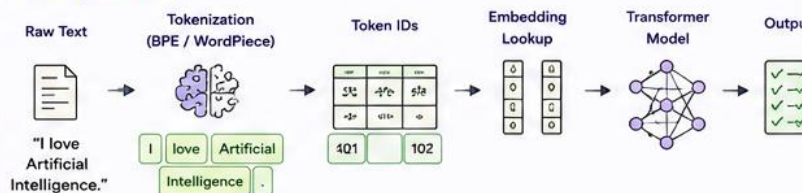


(## means continuation of previous token)

4. Comparison: BPE vs WordPiece

Aspect	BPE (used in GPT, Llama, RoBERTa)	WordPiece (used in BERT)
Method	Merge most frequent pairs	Merge based on best probability
Continuation Marker	No special marker	Uses ## for continuation
Example (playing)	play ing	play ##ing
Used In	GPT, Llama, RoBERTa, etc.	BERT, ALBERT, DistilBERT, etc.

5. Complete Flow



Key Takeaway

- Tokenization converts text to tokens.
- BPE merges most frequent pairs.
- WordPiece merges based on probability and uses ## for continuation.
- Subword tokenization helps models handle new and rare words efficiently.

- **Thank you**